

Week A. By Deng Yufan.

Prob 1.

a. Let X be random variable in $[0, 1]$. Prove if $E(X) = \epsilon$, then $P(X \geq \frac{\epsilon}{2}) \geq \frac{\epsilon}{2}$.

~~We have $P(X \geq \frac{\epsilon}{2})$~~

Let $Y = 1 - X$. We have $P(Y > 1 - \frac{\epsilon}{2}) = P(Y \geq 1 - \frac{\epsilon}{2}) = \frac{1 - \epsilon}{2} = \frac{1 - 2\epsilon}{2 - 2} = \frac{1 - 2\epsilon}{2 - 2} = 1 - \frac{\epsilon}{2}$
 $E(Y) = 1 - \epsilon$.

So $P(X \geq \frac{\epsilon}{2}) \geq \frac{\epsilon}{2}$.

b. Let $X \geq 0$. Prove $P(X = 0) \leq \frac{\text{Var}(X)}{E^2(X)}$.

~~Let $Y = X$ or $-X$, both are $\frac{1}{2}$ Probability then $E(Y) = 0$.~~

We have $P(|X - E(X)| \geq E(X)) \leq \frac{\text{Var}(X)}{E^2(X)}$ that is

$P(X=0) = P(X \leq E(X), |X - E(X)| \geq E(X)) \leq P(|X - E(X)| \geq E(X)) \leq \frac{\text{Var}(X)}{E^2(X)}$.

~~So $P(X=0)$~~

Prob 2.

Let $v_1, \dots, v_m \in [-1, 1]^n$ with equal prob. $w_i = \frac{v_i}{\sqrt{n}}$. Prove:

a. $i \neq j$: $P(|\langle w_i, w_j \rangle| \geq \delta) \leq \exp(-\Omega(\delta^2 n))$.

We have $w_{ik} \cdot w_{jk} \in [-\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}]$ both. let $u_k = w_{ik} \cdot w_{jk}$.

$$\begin{aligned} P(|\sum u_k| \geq \delta) &\leq 2 P(\sum u_k \geq \delta) \\ &= 2 P(\sum e^{\lambda u_k} \geq e^{\lambda \delta}) \\ &\leq 2 \frac{E(e^{\lambda \sum u_k})}{e^{\lambda \delta}} \end{aligned}$$

$$E(e^{\lambda \sum u_k}) = \frac{e^{\frac{\lambda^2}{2n}} + e^{-\frac{\lambda^2}{2n}}}{2} \leq e^{\frac{\lambda^2}{2n}} \text{ by Taylor.}$$

$$\text{Then } P(|\sum u_k| \geq \delta) \leq 2 \cdot \exp(\frac{\lambda^2}{2n} - \lambda \delta) = 2 \exp(-\frac{n \delta^2}{2})$$

when $\lambda = n\delta$.

b. $\exists m = \exp(\Omega(\delta^2 n))$ that $P(|\langle w_i, w_j \rangle| \leq \delta \forall i \neq j) \geq 0.99$.

We have $P(|\langle w_i, w_j \rangle| \leq \delta \forall i \neq j) \geq 1 - \binom{m}{2} P(|\langle w_1, w_2 \rangle| \geq \delta)$
~~when n is constant.~~

$$\begin{aligned} &\geq 1 - m(m-1) \exp(-\frac{n \delta^2}{2}) \\ \text{when } m &= 0.99 \exp(\frac{n \delta^2}{4}), \\ &\geq 0.99. \end{aligned}$$



Prob 3. Let G has n vertex. ^{undirected.} every edge has prob. p exist. Prove:

a. if $p = o(n^{-2/3})$, $\lim_{n \rightarrow \infty} P(G \text{ contains 4-clique}) = 0$.

$$P(G \text{ contains 4-clique}) \leq \binom{n}{4} p^6$$

$$= O(n^4) \cdot o(n^{-4}) = o(1).$$

b. if $p = \omega(n^{-2/3})$, $\lim_{n \rightarrow \infty} P(G \text{ don't contain 4-clique}) = 0$.

Let X be #4-clique. $E(X) = \binom{n}{4} p^6 = \omega(1)$. ~~then $E(X) = \omega(1)$.~~

$$\text{Var}(X) = E[(X - E(X))^2] = E(X^2) - E^2(X).$$

$$E(X^2) = \binom{n}{8} \binom{8}{4} p^{12} + \binom{n}{7} \binom{7}{4} \binom{4}{1} p^{12} + \binom{n}{6} \binom{6}{4} \binom{4}{2} p^{11} + \binom{n}{5} \binom{5}{4} \binom{4}{3} p^9 + \binom{n}{4} p^6.$$

$$\text{then } E(X^2) - E^2(X) = \binom{n}{8} \binom{8}{4} p^{12} (1-p) + \binom{n}{7} \binom{7}{4} \binom{4}{1} p^{12} (1-p^3) + \binom{n}{6} \binom{6}{4} \binom{4}{2} p^{11} (1-p^6) + \binom{n}{5} \binom{5}{4} \binom{4}{3} p^9 (1-p^3) + \binom{n}{4} p^6 (1-p^6).$$

$$= O(n^{-1/3}) + O(n^{-1}) + O(1) = O(1).$$

$$\text{We have } P(X=0) \leq \frac{\text{Var}(X)}{E^2(X)} = \frac{O(1)}{\omega(1)^2} = o(1).$$

